

midWife: Enabling Open-Ended Planning via Socratic Dialogue with Large Language Models

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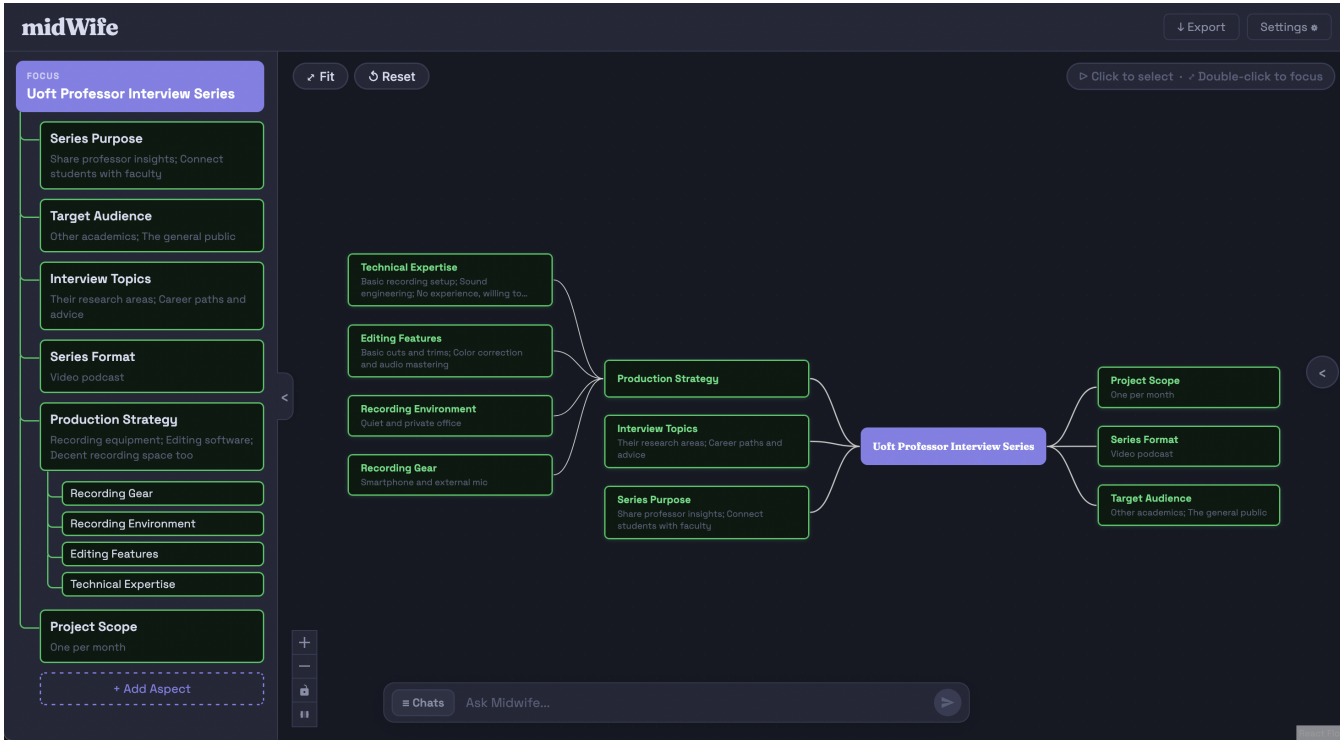


Figure 1: The MidWife discourse canvas. A user’s planning objective is decomposed into AI-generated Aspects — focused Socratic questions across key dimensions. Users answer via suggestion chips or free text; the tree expands recursively on elaboration. A per-node chat assistant (centre) guides the user to a concrete answer. The synthesis panel (right) distils the full discourse into actionable summaries adapted to the selected planning mode.

Abstract

Large Language Models (LLMs) have enabled users to engage in planning, reasoning and sensemaking with a “thought partner” of sorts, allowing open-ended discussions on nuanced subjects. This mirrors the idea of a *Socratic interlocutor* asking probing questions — serving to help users formalize their thinking, flesh out new ideas, and even find holes in their planning. However, most LLM platforms are limited to one-dimensional chat interfaces that cannot adjust to the scope of a dynamic, ever-growing discourse, thereby limiting the ability of a user to build a map of their thinking. This reflects an inherent limitation in the affordances offered by a linear conversational interface. We present *midWife*: an interactive

system that encourages thinking by iteratively soliciting the user’s intent, objective and/or nascent thoughts, probing with revelatory questions to concretize user-indicated notions, and visualizing this discourse as a map of thinking in the form of a node-link diagram connecting a central Thesis with its many Aspects. At each iteration, the user may (i) solicit more probing questions conditioned on a specific Aspect (if they are unsure of what to explore next); or (ii) manually add one or more Aspects at any point of the map (if they already know what to explore). Parallely, a Plan is generated by midWife in rich Markdown, updated incrementally to reflect the growing map of thinking. A persistent chat interface allows users to ask questions, request changes, and steer the Plan toward their focus areas.

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CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; *Natural language interfaces*; • **Computing methodologies** → Natural language processing.

Keywords

Socratic dialogue, discourse trees, structured thinking, LLM-assisted planning, mixed-initiative interaction, sensemaking

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1 Introduction

Planning complex, multi-dimensional tasks is an increasingly common yet poorly supported use case for large language models. Whether deciding how to change careers, organizing a major event, or working through a research hypothesis, users routinely turn to LLM-powered chat assistants as thinking partners [3]. Yet every major LLM interface organizes interaction as a flat, scrollable chat thread — a one-dimensional medium that cannot represent the branching, recursive structure of real deliberation [9]. A user who spends hours in a chat session planning a complex project has no spatial overview of what has been considered, no ability to navigate across parallel concerns, and no persistent artifact to return to and refine. Doing so with text-heavy media quickly becomes clunky; further, a user who wishes to explore a tangent cannot do so without polluting the shared context and potentially derailing the main thread of discussion. The *scope* of the discourse outgrows what the *interface* can hold.

The Socratic Method has long provided an answer to this kind of open-ended inquiry. Rather than providing answers, a skilled interlocutor asks focused questions that surface hidden assumptions, expose logical gaps, and map the conceptual space of a problem before any commitment is made [20, 24]. Socrates himself described this role as that of an “intellectual midwife”: whereas midwives help give birth to children, his questioning helped give birth to wisdom and ideas in others [17]. Applied to interactive systems, this lineage produced argument mapping tools rooted in Rittel and Kunz’s IBIS [8, 18], and more recently AI-powered dialectical systems [2, 23]. What remains missing is a single system that combines *spatial discourse structure* with *AI-as-Socratic-interlocutor* — one where questioning drives the growth of a persistent, navigable tree of thinking.

We present **midWife**¹: an interactive web application for open-ended planning via Socratic dialogue. Users state a planning objective and optionally select a *mode* (logistics, brainstorming, creative, decision, research, and more). midWife responds with a set of *Aspects* — targeted Socratic questions spanning key dimensions of the objective — rendered as nodes in a branching discourse tree. Answering an Aspect (via suggestion chips or free text) unlocks

Elaboration, which generates child Aspects conditioned on the parent’s answer and full ancestry context. A per-node chat assistant guides users through difficult questions, surfacing new Aspects or reframing questions as needed. As the tree grows, a *Synthesis Panel* distills the full discourse into structured, actionable output — timelines, task lists, open questions — adapted to the selected planning mode. The entire tree is a persistent, resumable artifact.

This paper makes the following contributions:

- **midWife**: a working prototype of LLM-powered Socratic planning via interactive discourse trees, built with a React + ReactFlow frontend and a FastAPI backend.
- A **mode-aware questioning engine** supporting nine distinct planning contexts, each modulating the AI’s Socratic questioning strategy.
- A **mixed-initiative interaction model** in which user and AI jointly construct, navigate, and refine the discourse tree [5].
- **Design insights** from iterative prototype development, including a proposed evaluation plan for future work.

2 Related Work

2.1 AI-Augmented Sensemaking and Planning Tools

A wave of systems published at UIST and CHI since 2022 have directly addressed the limitation of linear chatbot interfaces for complex information work. Sensecape [22] provides an infinite canvas where users interact with an LLM through multilevel abstraction, moving fluidly between information foraging and sensemaking loops; a within-subjects study found users explored significantly more topics and structured knowledge more effectively than with a ChatGPT-plus-canvas baseline. Selenite [12] solves the “cold-start” problem in unfamiliar decision domains by generating structured overviews of options and criteria upfront, dynamically contextualizing new information within those structures and accelerating information processing by 36% across three studies. Luminate [21] prevents premature convergence by first extracting key dimensions from a prompt, then generating responses that span the resulting design space; a study with 14 professional writers showed substantially increased divergent thinking. Graphologue [6] converts LLM responses into interactive node-link diagrams in real time, demonstrating that graphical, non-linear dialogue improves information parsing and exploration. Synergi [7] exemplifies mixed-initiative scholarly sensemaking: user-highlighted text seeds AI-generated hierarchical research threads grounded in citation context. Most closely aligned is the Human-AI Deliberation system of Ma et al. [13], which engages users in dimension-level opinion exchange and iterative discussion rather than presenting static recommendations — outperforming traditional explainable AI in fostering appropriate reliance.

Beyond the academic literature, a growing class of commercial mind-mapping and planning tools has begun integrating LLMs in this space. Tools such as Whimsical AI, Miro AI, and various Notion-based workflows typically offer one of two interaction patterns: (i) the user provides a short prompt or intent and the tool generates a complete mind-map or document in a single shot; or (ii) the user uploads an existing artefact — a PDF, transcript, or note — and the tool extracts a structured map from it. Neither pattern is interactive

¹The name is a direct tribute to the Socratic metaphor: an AI interlocutor that helps users bring their plans and ideas into being, much as Socrates helped his interlocutors give birth to wisdom.

in the deliberative sense. In the first, the AI does all the thinking and the user is a passive recipient; in the second, the AI merely reorganises existing material. In both cases, the user’s agency is limited to accepting, rejecting, or lightly editing the AI’s output. The co-creative, back-and-forth quality of genuine planning — the iterative questioning, the gradual crystallization of vague intent into concrete decisions — is absent. As a consequence, users report lower sense of ownership over the produced artefact. midWife’s design is explicitly a response to this pattern: its core invariant is that *the AI never makes decisions on the user’s behalf*.

A common thread across both academic systems and commercial tools is that AI produces *content*: information, structure, synthesis. It organises and generates; it does not interrogate. midWife’s core departure is that its primary AI role is to *ask*, not to tell.

2.2 Socratic and Dialectical AI Interaction

Tanprasert et al. [23] demonstrated in a landmark CHI 2024 Best Paper that LLM-powered debate chatbots with outgroup identity and persuasive rhetoric most effectively induced critical thinking — interpretation, analysis, and self-regulation — in users engaging with online video content. Chiang et al. [2] showed that interactive devil’s advocate agents that challenge AI recommendations improved appropriate reliance and decision accuracy by catalyzing extended discussion. Lim et al. [10] combined fallacy detection with Socratic questioning in a browser extension, reporting that it encourages attentiveness and poses thought-provoking questions in evaluation of potential misinformation. At the model level, SocraticLM [11] fine-tunes LLMs for Socratic teaching through a multi-agent pipeline, producing a dataset of 35K Socratic dialogues — validating that LLMs can reliably ask guiding questions rather than give answers at scale. Chang [1] engineers ten Socratic strategies from Plato’s dialogues (elenchus, maieutics, dialectic, counterfactual reasoning) into prompt templates, demonstrating improved output correctness and more critically examined responses. Lee et al. [9] provide an empirical motivation for midWife’s design: surveying 319 knowledge workers, they find higher AI confidence correlates with *less* critical engagement by users — underscoring the need for AI systems that provoke thought rather than satisfy it. Collins et al. [3] frame the aspirational target as AI “thought partners”: reasonable, insightful, trustworthy collaborators that think *with* humans.

A shared limitation across this stream is structural: these systems operate within flat, linear chat interfaces. The dialogue may be dialectical, but the representation is one-dimensional. midWife embeds Socratic questioning inside a persistent, spatial discourse tree — giving the dialogue a structure that users can navigate, revisit, and build upon.

2.3 Non-Linear Conversation and Argumentation

The intellectual lineage of structured discourse tools begins with Kunz and Rittel’s IBIS [8], which defined a grammar for structuring complex policy deliberation around Issues (questions), Positions (answers), and Arguments (pros/cons). Rittel and Webber’s complementary “wicked problems” work [18] established why such structure is necessary: ill-structured problems demand argumentative,

deliberative processes rather than algorithmic resolution. Toulmin’s argument model [24] contributed a richer grammar (claim, grounds, warrant, backing, qualifier, rebuttal) that underpins virtually all subsequent computational argumentation work. Conklin and Begeman’s gIBIS [4] realized IBIS graphically, demonstrating that typed structure eliminates unconstructive conversational moves; Compendium extended this into a professional dialogue-mapping practice. MacLean et al.’s QOC notation [14] showed a structurally isomorphic pattern (Questions, Options, Criteria) for design rationale, confirming the generality of branching deliberation structures. Cognitive grounding comes from Russell et al.’s sensemaking cost model [19] and Pirolli and Card’s dual-loop foraging-sensemaking framework [16]: explicit tree representations reduce the cognitive cost of organizing complex thought by externalising working memory.

Recent work has begun augmenting this lineage with LLMs. VISAR [26] supports argumentative writing through hierarchical goal brainstorming and LLM-powered “argumentation sparks” (counterarguments, evidence, logical weakness detection) within a tree-structured outline. AMQuestioner [15] automatically extracts argument maps from online discussions and presents Socratic questions to guide users in critically exploring claims; a study of 24 participants showed improved critical thinking skills and open-mindedness. Wang et al. [25] provide direct empirical support for midWife’s design choice: node-tree interfaces outperformed chatbots specifically on *decision-making* and *brainstorming* tasks, with 90% of node-tree users finding the interface easy to navigate. Horvitz’s mixed-initiative principles [5] supply the interaction design framework for managing when the AI should take initiative versus deferring to the user.

Despite this progress, these systems use AI primarily for *content generation*: surfacing counterarguments, labelling argument types, retrieving evidence. None positions the AI as a *Socratic facilitator* whose primary goal is to help the user think more deeply through structured questioning.

The three threads described above are active and productive, yet no existing system integrates all three. Table 1 characterizes the landscape. Sensemaking tools provide spatial or hierarchical structure but the AI generates *content*, not questions. Dialectical AI systems challenge and interrogate but do so within flat, linear interfaces. Argumentation tools provide structured discourse representations but use AI only for content generation, not Socratic facilitation. midWife’s contribution is the synthesis: a tree-structured dialogue interface where the AI acts as a Socratic interlocutor, combining the spatial structure of Sensecape and Graphologue, the dialectical AI role of Human-AI Deliberation and the debate chatbot literature, and the argument grammar of IBIS and Toulmin — extended with mode-aware questioning across nine distinct planning contexts.

3 MIDWIFE

3.1 Motivating Scenario

Consider Priya, a software engineer contemplating a move abroad for a new career opportunity. She opens a chat assistant and types her situation — and immediately faces the core problem: a good answer requires exploring dozens of interconnected dimensions. What is her financial runway? Does she have the right visa? What

Table 1: Positioning of midWife relative to representative prior systems. ✓ = fully present, ~ = partial, × = absent.

System	Spatial tree	Socratic AI	Mode-aware
Sensecape [22]	✓	×	×
Human-AI Deliberation [13]	×	~	×
Debate Chatbots [23]	×	✓	×
VISAR [26]	✓	~	×
AMQuestioner [15]	✓	~	×
midWife (ours)	✓	✓	✓

happens to her current apartment and social network? What are her priorities — career growth, lifestyle, proximity to family? A linear chat thread surfaces some of these concerns, but has no way to hold them simultaneously. Exploring one thread buries earlier ones. A good follow-up about housing arrives twenty messages after the topic was first raised; the context is effectively stale. Worse, when Priya wants to take a quick tangent — say, a digression about her lease terms — she cannot do so without polluting the shared context and derailing the main thread.

With midWife, Priya types: “I’m thinking of moving to Berlin for a job opportunity” and selects the *Decision* mode. midWife generates five Aspects: *Finances*, *Visa & Logistics*, *Career Fit*, *Lifestyle*, and *Family & Social*. Each appears as a node on a radial discourse canvas. She clicks *Finances*: midWife asks, “How long could you sustain yourself financially if you spent a few months settling in before starting?” She types her answer. She clicks *Elaborate* — midWife generates three child Aspects: *Cost of Living*, *Savings Buffer*, and *Currency Risk*. She navigates back up and clicks *Career Fit*, opening the per-node chat assistant. She is unsure how to think about this angle; the assistant asks, “What would you want to have achieved professionally in two years that your current role does not allow?” — guiding her to a concrete answer without providing one. After exploring four of the five top-level Aspects, she clicks *Synthesize*. midWife produces a structured Decision Panel: an overview, a list of open questions, a recommendation summary, and actionable next steps. Priya did not *receive* a plan — she *built* one.

3.2 Design Goals

D1 — Visualize Discourse as a Dynamic Map of Thinking. A linear chat transcript cannot externalize the structure of multi-dimensional thinking. Our first goal was to find a better *medium*: one in which the growing discourse is visible as a spatial artefact the user can navigate. Given the user’s planning objective, midWife polls the LLM for the five or six most pertinent *Aspects* worth exploring. Each becomes a node in a radial graph, with the user’s Thesis (their original intent or objective) at the centre and Aspects as immediate children. As the user answers questions and elaborates branches, the tree grows outward, accumulating a visual record of what has been considered, decided, and left open. A Navigator Pane on the left of the interface allows users to add, edit, move, and delete nodes at any time, ensuring the map remains theirs to shape.

D2 — Probe, Rather Than Answer, on the User’s Behalf. Many contemporary mind-mapping and planning tools offload the thinking away from the user entirely: the user provides an initial

intent and is presented with a complete plan, map, or document generated by the AI. While efficient, this pattern significantly diminishes the user’s *sense of ownership* over the produced artefact — the final outcome does not carry their fingerprint. We take a fundamentally different approach. midWife uses the LLM exclusively to: (i) ask probing questions; (ii) uncover hidden assumptions; (iii) help the user verbalize inchoate ideas through focused follow-ups; and (iv) offer concrete suggestions only when the user is navigating an unfamiliar domain. The *onus* of deciding, committing, and directing always rests with the user. This design choice is testable: we expect users to report higher agency, ownership, and satisfaction with the output than users of single-shot generation tools — a hypothesis for future evaluation.

D3 — Model Thinking as a Mixed-Initiative Dialogue. Effective deliberation is neither fully user-driven nor fully AI-driven; it is a back-and-forth in which each party contributes what the other cannot easily do alone. We therefore designed midWife so that *every significant action* can be initiated by either the user or the AI. New Aspects can be AI-generated (via elaboration) or user-created (via the Navigator Pane or manual node creation). Questions can be AI-authored or user-edited. The Synthesis Panel is initially generated by the AI based on the completed discourse, but the user retains full executive control: they can edit, delete, or redirect any section of the plan via natural language chat. This mirrors the core principles articulated by Horvitz [5]: the system should scope its initiative to the user’s current context, offer safety nets (e.g. editable suggestions), and time AI interventions to moments when the user is most likely to benefit from them.

3.3 The Discourse Tree Model

Each session in midWife is called a *Discourse* — a structured planning session centred around a *Thesis*: the user’s objective, intent, topic of interest, or open question. Before the Discourse begins, the user specifies a *Mode of Help*: one or more of Logistics, Brainstorm, Creative, Problem-Solving, Decision, Research, Reflect, Set Goals, or Learn, which shapes the AI’s questioning strategy throughout.

The fundamental unit of the Discourse is an *Aspect*: a dimension of the Thesis that warrants exploration. Each Aspect is a node in the tree carrying four fields: (i) a short *label* (at most three words, e.g. *Budget*, *Career Fit*); (ii) a *Socratic question* — a focused, beginner-friendly prompt to help the user think through that dimension; (iii) a set of three to five pre-generated *suggestion chips*; and (iv) the user’s *answer*, once provided. Answering an Aspect unlocks *Elaboration*: midWife generates child Aspects whose questions are conditioned on the full *ancestry path* — every aspect/question/answer triple from the root down to the current node. This ensures deeper questions are meaningfully scoped to what the user has already decided.

Recontextualization addresses a subtle issue that emerges as the discourse deepens: an ancestor’s short label may no longer accurately reflect the scope of its subtree. The system periodically re-evaluates ancestor labels against their children’s content and proposes updated labels, keeping the map legible as the discourse grows.

3.4 User Interface & Features

After the user submits their Thesis and Mode of Help, the interface transitions to the main Discourse page, which consists of three primary areas: (i) the **Navigator Pane** on the left, showing a structured list of all *Aspects* in the current Discourse, allowing users to add, edit, move, and delete nodes at any time; (ii) the **Discourse Canvas** at the centre, a node-link diagram rendering the Thesis and its Aspect nodes radially, updated in real time as the Discourse grows; and (iii) the **Plan Pane** on the right, displaying a rich Markdown summary of action items, timelines, and open questions synthesised from the current state of the Discourse.

3.4.1 Interview & Answering. For each *Aspect* in the Discourse, midWife engages the user in an **Interview**: a focused, one-at-a-time question session. The user is presented with the Socratic question for the selected Aspect alongside up to five pre-filled suggestion chips. They may select one or more chips, add detail in a free-text box, or type a fully custom answer. If they are uncertain how to approach the question, they may choose *Chat about this*, which opens the **Chat Interface** anchored to that specific node. Here, the LLM helps the user articulate their requirements, resolve ambiguity, and arrive at either a pre-specified option or a custom answer of their own. Critically, the chat assistant does not answer the question for the user: it asks, clarifies, and guides until the user reaches their own conclusion.

3.4.2 Mode-Aware Questioning Engine. midWife supports nine planning modes — Logistics, Brainstorm, Creative, Problem-Solving, Decision, Research, Reflect, Set Goals, and Learn — each of which injects a distinct instruction paragraph into the LLM’s system prompt, redefining what constitutes a relevant Socratic question for that context. Users may select a single mode or combine multiple for hybrid sessions.

3.4.3 Navigator Pane. The Navigator Pane provides a structured list view of the Discourse, organised around the concept of a **Focus Node**. By default the root Thesis is the Focus Node, and the Pane displays it alongside up to its grandchildren — giving users a bounded, readable slice of the tree at any one time. Users may double-click any Aspect to re-focus: the Canvas recentres on that node and the Pane narrows to its local subtree, allowing deep branches to be explored without the surrounding tree becoming a distraction.

Within the Navigator Pane, users can add new Aspect nodes to any existing node or to the Thesis itself, edit labels, delete nodes, and move a node to become the child of a different parent if they feel the structure better reflects their thinking. Nodes can also be expanded by the AI: each Aspect card carries an **Explore** button that instructs midWife to generate additional directions to investigate under that Aspect, surfacing branches the user may not have considered.

3.4.4 Plan Pane. While the Discourse Canvas captures the *structure* of a user’s thinking, the Plan Pane synthesises its *implications*. It surfaces information that the canvas does not directly represent: concrete action items and todos derived from answered Aspects, a timeline if the user’s objective has a temporal dimension, open questions that remain unresolved in the tree, and risks or considerations that are still unaddressed. All of this is rendered as a rich Markdown document, updated on demand as the Discourse grows.

Users steer the Plan through the chat interface at the bottom of the Discourse Canvas. Each requested edit — rephrasing a section, adding a new consideration, restructuring the timeline — is first shown as a **diff preview**; the user can accept the proposed changes or dismiss them and keep the current Plan. This accept/reject flow ensures the Plan remains under the user’s control rather than being silently overwritten by the AI.

3.5 Mixed-Initiative Interaction

midWife distributes initiative between user and AI across every layer of the interaction. On the AI side: the system generates the initial set of Aspects on session creation; pre-generates suggestion chips for each node; proposes ancestor relabelling via recontextualization; and synthesises the full discourse into the Plan Pane on demand. On the user side: the user answers via chips or free text; selects which branches to elaborate; manually adds, moves, deletes, and relabels nodes via the Navigator Pane; opens the per-node chat for any question they find difficult; and edits the Plan Pane via natural language directives in the chat interface.

A key design decision is scoping the chat assistant to individual nodes rather than providing a single global chatbot. Each chat session is initialised with the full context of the selected node — its label, Socratic question, and the complete ancestry path of prior answers — and is given one directive: help the user reach a concrete answer to *this* question. The assistant will reframe the question if the user finds it irrelevant, surface a new Aspect if a genuinely new concern emerges, and extract a suggested answer as soon as the conversation converges. Each node’s chat history is preserved independently, allowing users to track the reasoning behind any answer and return to prior conversations without losing context. This scoping prevents context bleed across unrelated branches and ensures the chat always serves the discourse tree rather than replacing it.

4 Implementation

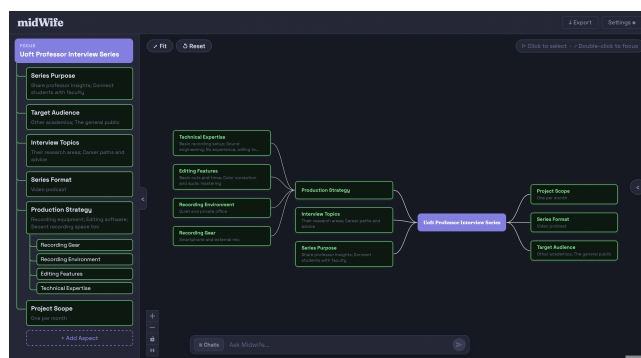


Figure 2: MidWife system architecture. The React frontend communicates with a FastAPI backend over a REST API. Claude Haiku serves as the primary LLM; Gemini models provide a cascading fallback. Session state is maintained server-side and synced to browser localStorage, enabling resumable sessions without a database.

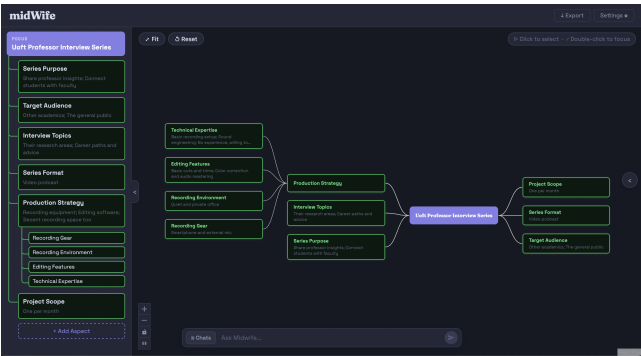


Figure 3: MidWife in use. (a) A planning session in progress: answered Aspect nodes (dark fill) branch into revealed children; the selected node's chat assistant is open. (b) The synthesis panel generated from the completed discourse, showing mode-specific tabs (Overview, Timeline, Tasks & Resources, Open Questions).

midWife is built on a React + Vite frontend and a FastAPI backend (Figure 2). The discourse canvas is implemented with ReactFlow, which provides native pan, zoom, and node-selection at no interaction cost to the user. Tree layout is handled by Dagre, which positions nodes automatically as the tree grows — left and right children are assigned based on edge source handle, ensuring the mindmap-style branching remains spatially consistent regardless of tree depth or width. Layout constants (rank separation, node separation) were tuned iteratively so that a fully elaborated tree with 20–30 nodes remains readable without manual rearrangement.

The AI pipeline uses Claude Haiku 4.5 as the primary model, chosen for its low latency and reliable structured-JSON output — all Aspect labels, Socratic questions, and suggestion chips arrive in a single round-trip, keeping the interview feel responsive. A cascading Gemini fallback (2.0 Flash → 2.5 Flash → 3.1 Flash-lite) provides graceful degradation under rate limits, transparent to the user. Three prompt engineering decisions directly shape the interaction quality. First, *covered_aspects deduplication*: all existing node labels are injected into each generation call as a “do not repeat” constraint, so users never encounter a duplicate or near-duplicate Aspect as the tree grows. Second, *context_path injection*: the full ancestry of label–question–answer triples is included in every elaboration prompt, causing questions to grow progressively more specific with depth — an effect users perceive as the system tracking where they are in their thinking. Third, *mode-specific system prompts*: each of the nine planning modes installs a different questioning persona and Aspect taxonomy (e.g. Research mode produces hypothesis, evidence, and gap Aspects; Reflect mode produces introspective, first-person questions), so the same tree interaction model feels meaningfully different across planning contexts. Finally, when a user manually adds a node by specifying only a label, the backend’s generate_aspects endpoint synthesises a Socratic question from that label and its ancestry context, ensuring user-authored nodes are indistinguishable from AI-generated ones in the interview flow.

5 Discussion

5.1 Design Insights

TODO: Write design insights.

5.2 Limitations

TODO: Write limitations.

5.3 Future Evaluation

TODO: Write future evaluation subsection.

5.4 Future Directions

TODO: Write future directions.

6 Conclusion

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